Hochschule
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Master Thesis Defense

Computational Engineering (M.Sc.)

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Design and Implementation of a Real-Time Impedance Controller for a 7-DOF Collaborative Robot

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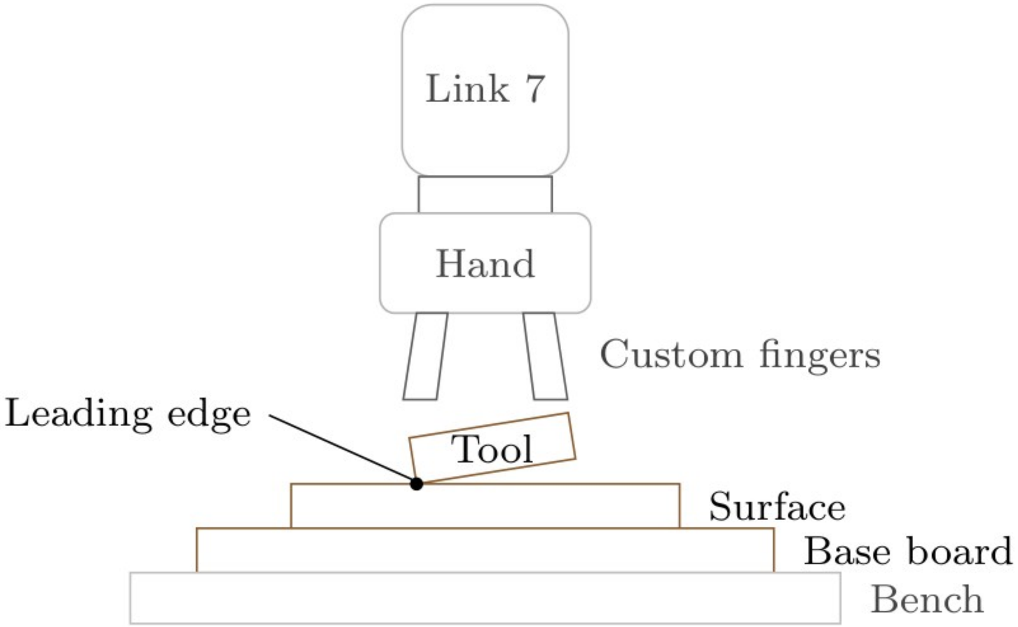
Motivation



Problem

- Tilted tool → edge-first contact
- Rigid tracking resists corrective rotation

Main thesis question

- **How do stiffness and the compliance centre affect passive alignment?**

Idea

- Apply the press; let contact rotate the tool

Overview

- Cartesian impedance
- Virtual centre of compliance
- Experiments and results
- Conclusion

Cartesian impedance

Translational spring–damper

$$f = K_p e_p + D_p(\dot{p}_d - \dot{p}_{\mathrm{EE}})$$

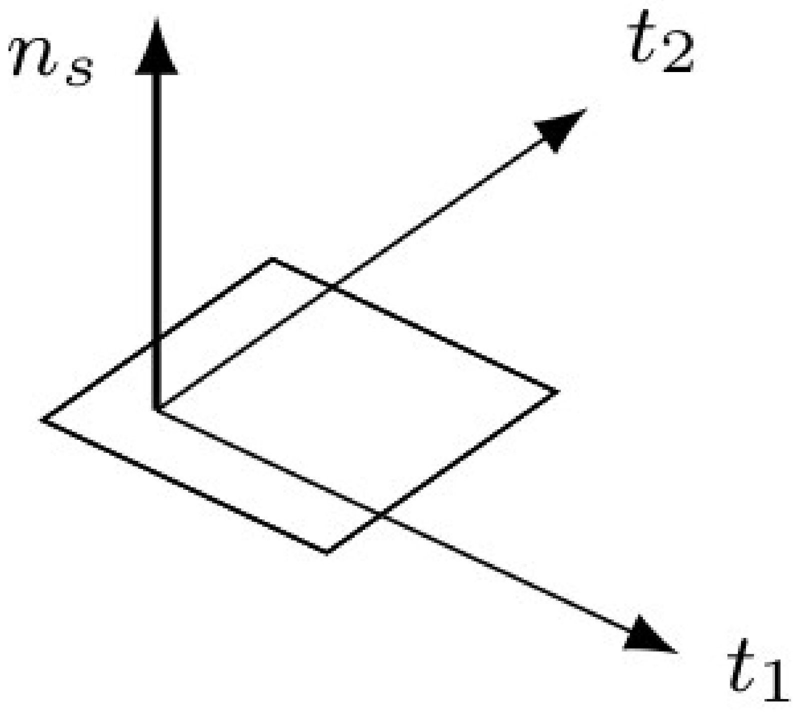
Rotational spring–damper

$$m = K_R e_R + D_R(\omega_d - \omega_{\mathrm{EE}})$$

- Stiffness resists pose errors
- Damping reduces oscillation
- Compliant contact instead of rigid tracking

- $f,\ m$ commanded force / moment
- $e_p,\ e_R$ position / orientation errors
- $K_p,\ K_R$ translational / rotational stiffness
- $D_p,\ D_R$ translational / rotational damping
- $\dot{p},\ \omega$ linear / angular velocity
- $d,\ \mathrm{EE}$ desired / measured end effector

Surface-relative compliance



- n_s outward surface normal
- $t_1,\ t_2$ two perpendicular surface tangents

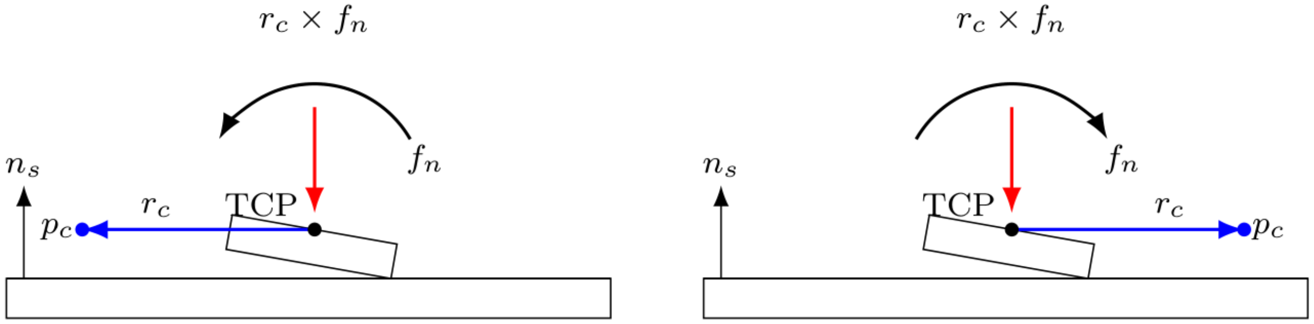
Translation

- Softer normal pressing
- Stiffer motion along the surface

Rotation

- Softer tilting about the tangents
- Stiffer turning about the normal

Centre of compliance (CoC)



(a) Selected displacement (b) Reversed displacement

- Tangential shift → preferred rotation
- Reversed shift → reversed additional moment

$$m = m_R + r_c \times f$$

- **TCP** tool centre point; controlled reference
- p_c virtual compliance centre
- r_c displacement from TCP to centre
- f_n normal part of commanded force
- m_R rotational spring–damper moment

Experimental procedure

Tool orientation → **Surface approach** → **Contact establishment** → **Pre-grinding hold**

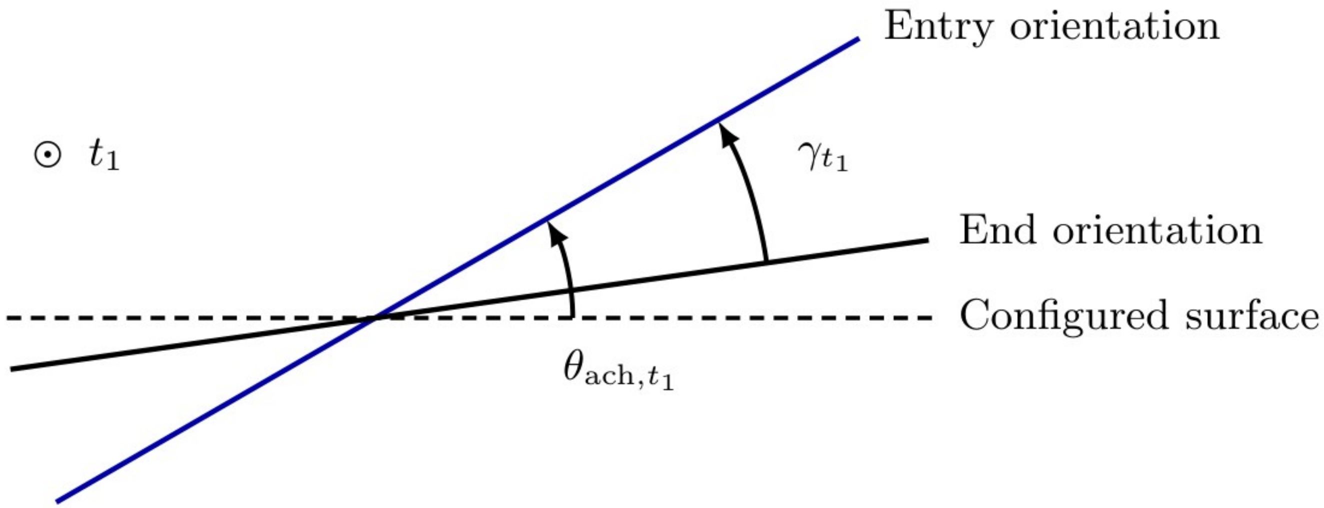
Contact study

- Rectangular tool: 40 × 120 mm
- Five-second contact phase
- Fixed entry orientation reference

Controlled comparison

- A: centre at the TCP
- B / C: rotational / translational stiffness
- D: tangential compliance-centre position
- 57 trials; three repetitions per setting

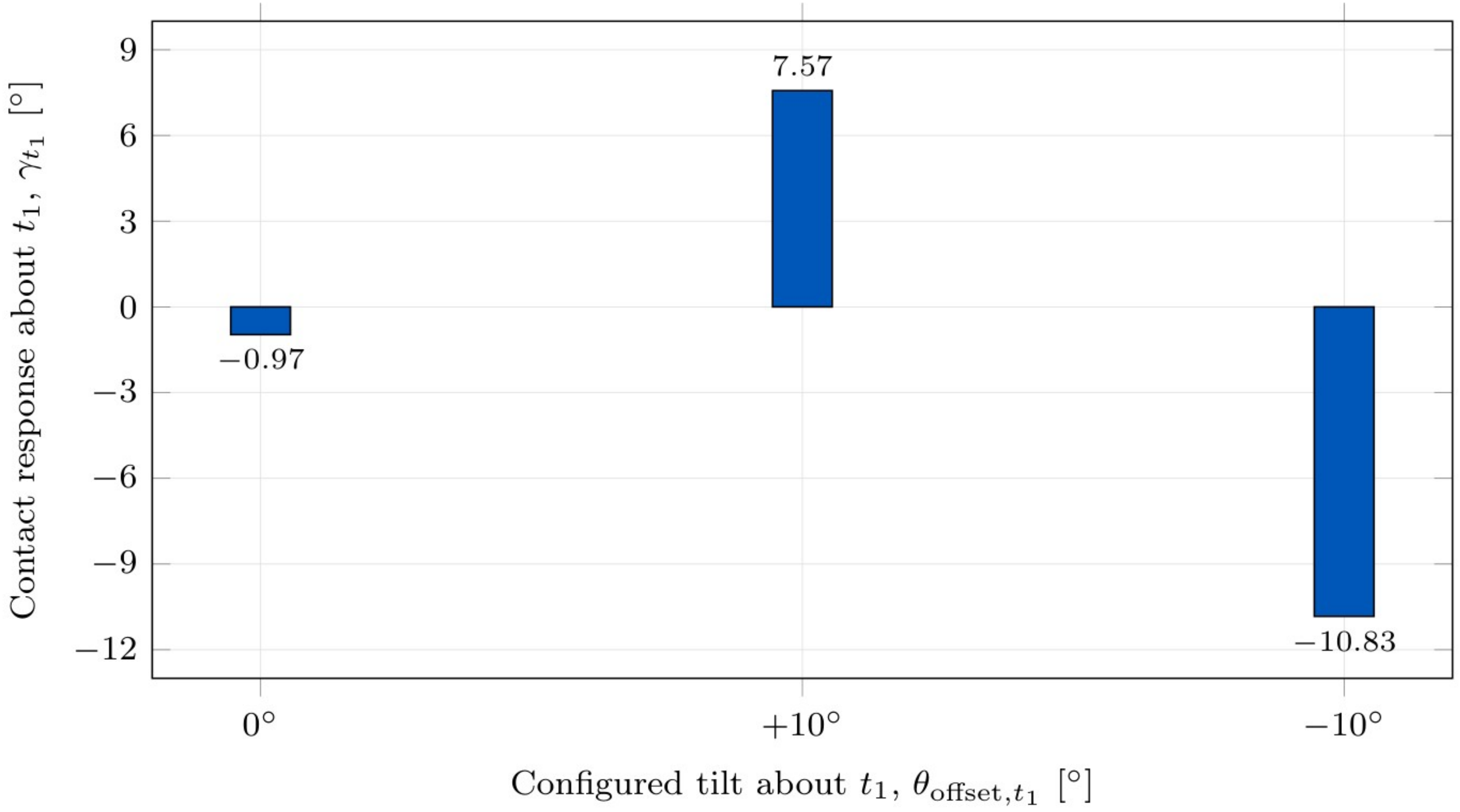
Angular quantities



- $\theta_{\mathrm{offset},t_1}$ commanded tilt before contact
- $\theta_{\mathrm{ach},t_1}$ achieved tilt at contact entry
- γ_{t_1} signed contact response about the first tangent

- Alignment: response has the same sign as the entry tilt
- Angles inferred from measured end-effector orientation

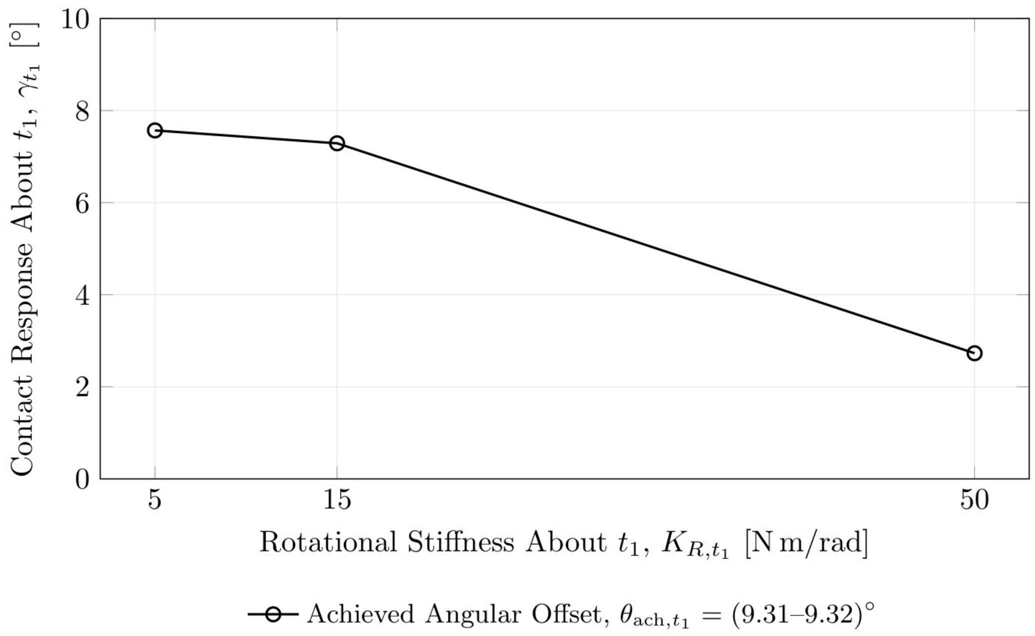
Baseline contact response



- Centre of compliance at the TCP
- Alignment response for both tilt signs
- Unequal response magnitudes

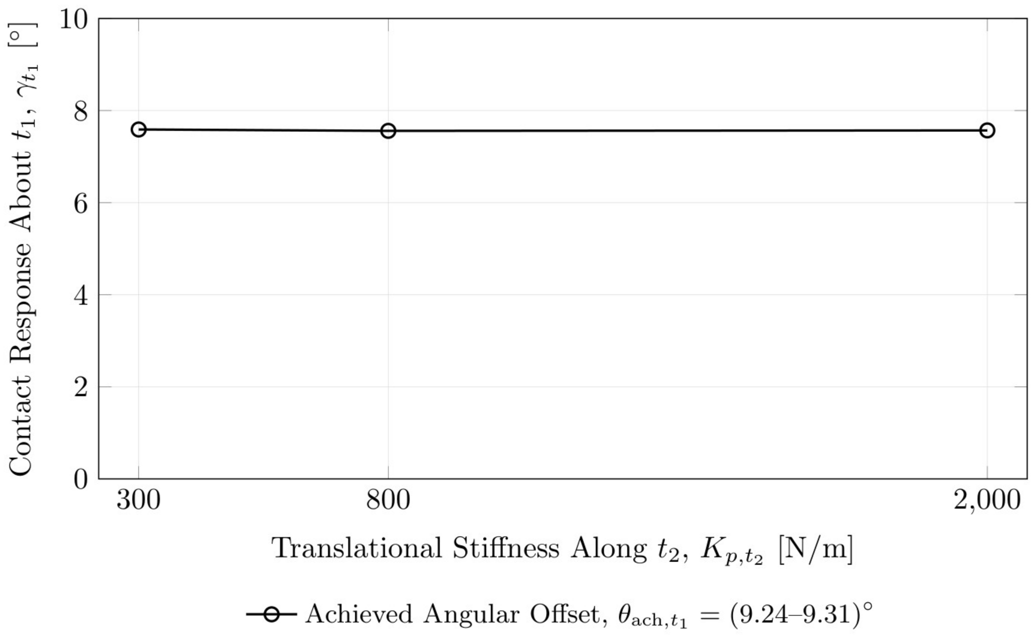
Effect of stiffness

- K_{R,t_1} resistance to tilting about the first tangent



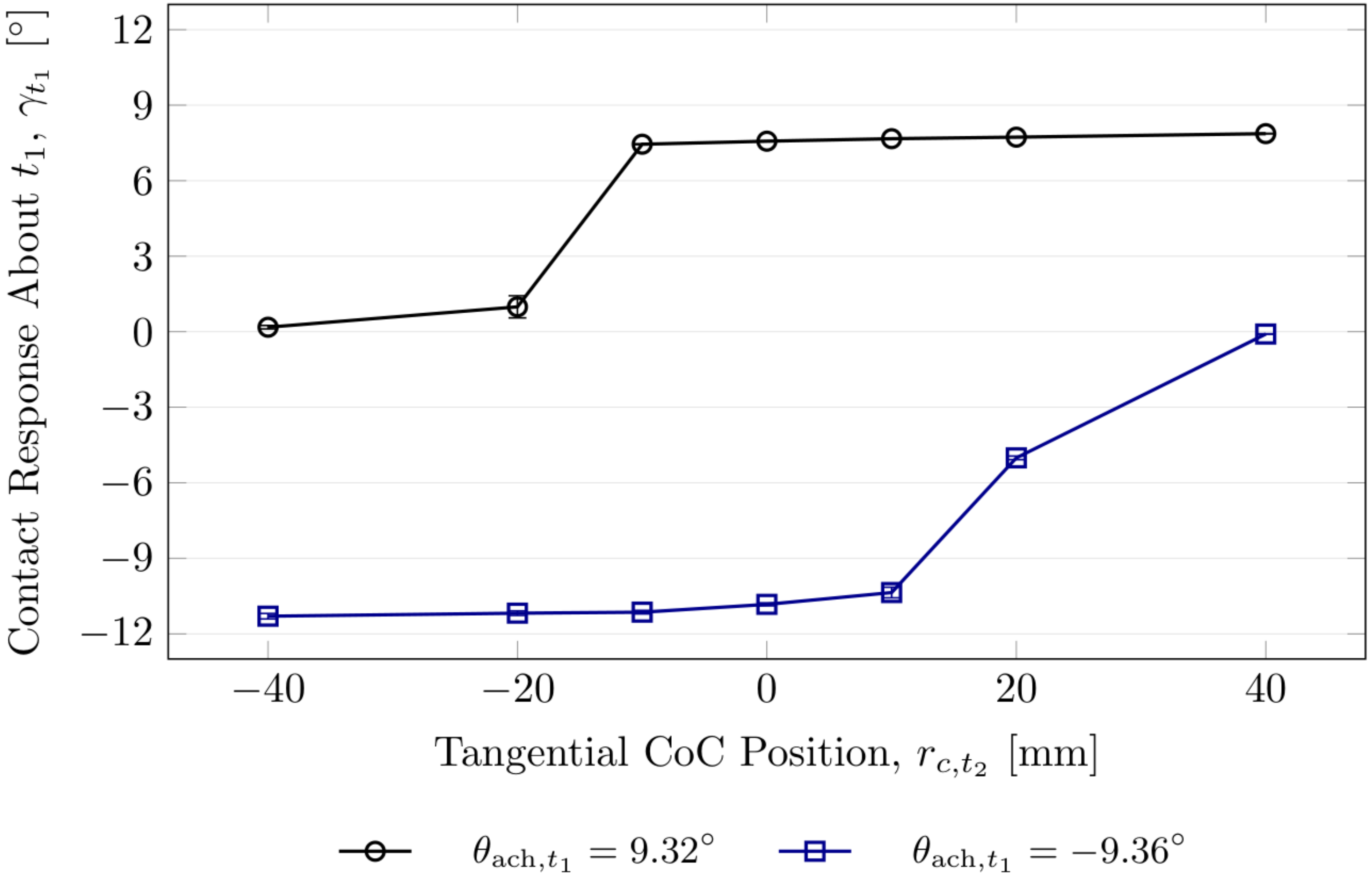
- **Rotational stiffness: 63.9% lower response**

- K_{p,t_2} resistance to sliding along the second tangent



- **Tangential stiffness: 0.4% response span**

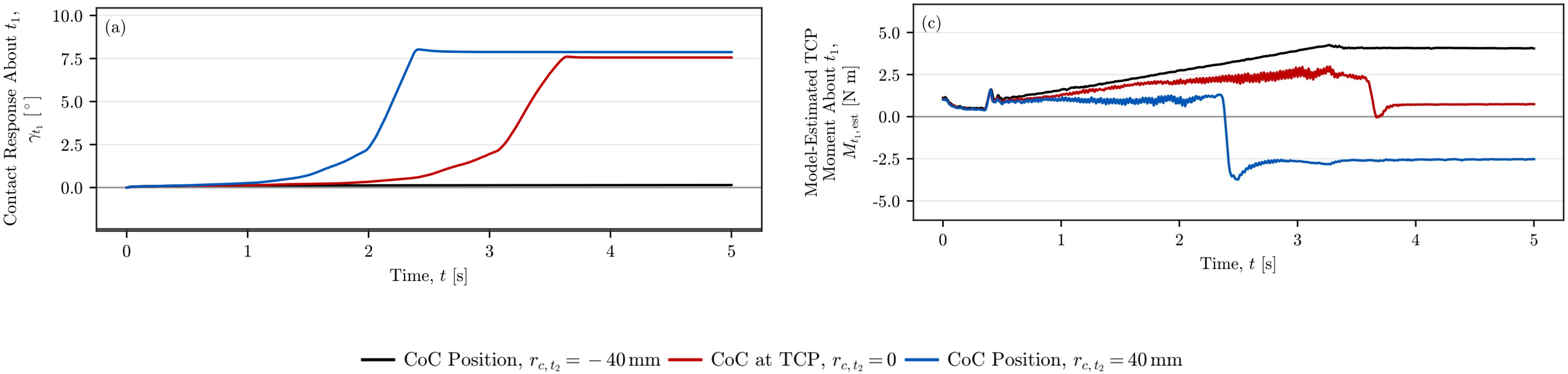
Effect of CoC position



- r_{c,t_2} CoC displacement along the second tangent
- Selected side: about 4% higher response
- Opposite side: 98–99% lower response
- Required side reverses with the entry tilt

Contact response and interaction moment

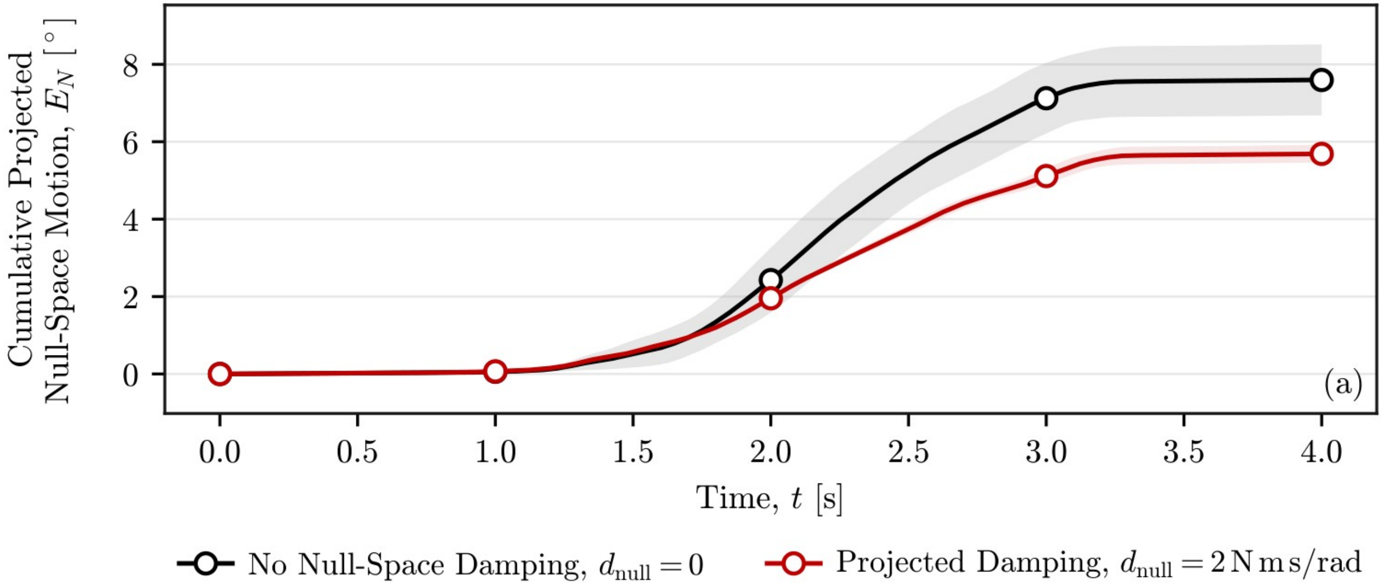
- t elapsed contact time [s]
- $M_{t_1,\mathrm{est}}$ model-estimated moment about the first tangent



- Similar normal-force levels across the three positions
- Moment reversal; selected side reaches its steady response about 1.3 s earlier

Secondary study: null-space motion

- E_N cumulative projected joint motion
- d_{null} null-space damping coefficient



Null space

- Extra joint motion while the tool pose is held
- Damping: about 25% less total motion
- Conditioning: almost zero final drift
- Stronger conditioning: more back-and-forth motion

Scope and next steps

Scope

- One tool, plane and robot configuration
- Tool angle inferred from robot pose
- Evaluation focused on contact entry

Next steps

- Independent tool-angle measurement
- Select the CoC from the entry tilt
- Return to the TCP; test sustained grinding

Conclusion

- Passive alignment through Cartesian impedance
- Rotational stiffness sets response magnitude
- CoC displacement creates a preferred direction
- Adaptive displacement: next step to evaluate

Thank you

Questions and discussion